Fiscal Year:	FY 2020	**Task Last Updated:**	FY 02/18/2020
PI Name:	Roberts, Donna M.D.		
Project Title:	Human Cerebral Vascular Autoregulation and Venous Outflow in Response to Microgravity-Induced Cephalad Fluid Redistribution		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	robertdr@musc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	843-792-4932
Organization Name:	Medical University of South Carolina		
PI Address 1:	Department of Radiology		
PI Address 2:	169 Ashley Avenue, POB 250322		
PI Web Page:			
City:	Charleston	**State:**	SC
Zip Code:	29425-8905	**Congressional District:**	6
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	05/16/2013	**End Date:**	09/30/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2020 per PI (Ed., 9/30/19) NOTE: End date changed to 9/30/2019 per NSSC information (Ed., 2/26/19)		
Key Personnel Changes/Previous PI:	February 2018 report: added Dr. Bloomberg as Co-Investigator.		
COI Name (Institution):	Brown, Truman Ph.D. (Medical University of South Carolina) George, Mark S M.D. (Medical University of South Carolina) Bloomberg, Jacob Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	NNX13AJ92G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE this is continuation of "Human Cerebral Vascular Autoregulation and Venous Outflow in Response to Microgravity-Induced Cephalad Fluid Redistribution" grant #NNX13AC32G, which ended 5/15/2013. According to the NASA Human Research Roadmap (Risk of Microgravity-Induced Visual Impairment/Intracranial Pressure; Gaps VIIP2, VIIP4, VIIP5, VIIP8) and further expounded upon in the Visual Impairment Intracranial Pressure (ICP) Summit Report (May 2011), an essential gap exists in our knowledge of intracranial pressure regulation in spaceflight [Ed. note September 2019: Risk changed in November 2017 to Spaceflight Associated Neuro-ocular Syndrome (SANS); Gaps also changed to SANS1, SANS3, SANS12, SANS13--seehttps://]. This is exemplified by the fact that there are few published studies investigating the structural and functional changes occurring as a result of long-term spaceflight and there is almost no data concerning cerebral autoregulation in astronauts following return from space. Only two published studies exist investigating cerebral autoregulation in astronauts using an indirect method for measuring cerebral blood flow (CBF) at the level of the major arteries of the circle of Willis. Thus, although cerebrovascular hemodynamics plays a crucial role in ICP regulation and also in the response to orthostatic stress, we have almost no knowledge of the state of cerebral autoregulatory mechanisms during spaceflight. The operational relevance of this gap according to the NASA Human Research Roadmap is that elevated ICP and likely associated vision disturbances due to microgravity exposure are serious health risks for astronauts. Furthermore, this is an even greater concern for potential longer duration missions such as to the moon or Mars where decreased vision may impact mission performance and may even require early mission termination. Permanent visual losses may result in lifetime disability issues. Almost certainly, during long-term flights, as the hydrostatic pressure gradients due to gravity are abolished with resultant large cephalad fluid shifts, an intracranial adaptive response occurs. Thus direct measurements of cerebral fluid shifts and cerebral hemodynamics as proposed in this study will allow us to more fully characterize the intracranial compartment guiding the development of preventive measures, such as specific pharmacological countermeasures, to combat the development and sequelae of increased ICP during future missions. In order to fully characterize intracranial adaptation to long-term microgravity, we will use magnetic resonance imaging (MRI) which is the terrestrial gold standard. MRI allows direct visualization and functional characterization of the intracranial contents. Indeed, MR imaging is the clinical standard of care in evaluating patients with idiopathic intracranial hypertension. We will use a suite of advanced MRI sequences to: (1.) characterize weightlessness-induced intracranial compartmental fluid volume changes at the macroscopic (brain, cerebrospinal fluid-CSF) and vascular (arterial and venous) levels following adaptation to long-term spaceflight and (2.) assess for potential resultant hemodynamic changes (cerebral perfusion) in the brain. This proposal represents the most relevant and advanced suite of MRI techniques for the characterization of intracranial compartmental fluid and hemodynamic changes induced by long-term microgravity. The specific sequences will be carried out by the recognized experts in each area and in most cases, the developer of the specific technique. It is imperative to obtain this data in order to have an understanding of what is happening to the brain and orbits while living and working in microgravity. We are only now realizing the consequences of this lack of knowledge. This data will serve as the basis for the development of fundamental theories describing intracranial adaptation to microgravity. It may help establish a risk stratification strategy based on intracranial anatomy, dural venous sinus and jugular vessel outflow, and cerebral vascular autoregulation. Importantly, it will serve as a basis for interpretation of in-flight data collected via other techniques such as ultrasound, the primary in-flight imaging modality.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Idiopathic intracranial hypertension, or pseudotumor cerebri, is a poorly understood clinical condition of unknown etiology. It has a reported incidence of between 1 and 5 cases per 100,000 people in the general population, although the true incidence may be larger. There is a strong female predominance. Patients typically present with disabling headaches and vision changes. In more extreme cases, injury to the optic nerve can occur and patients can experience vision loss. Upon testing, these patients have increased intracranial pressure and swelling of the optic nerves. Similar symptoms can occur in astronauts following long-term spaceflight. This study will provide a unique opportunity to investigate the underlying pathophysiology of abnormal intracranial pressure regulation which will be applicable to those who suffer from this Earth-based condition. This research may also shed light on the pathophysiology of other Earth-based disease processes associated with abnormal cerebrospinal fluid (CSF) flow or venous outflow obstruction such as obstructive hydrocephalus, normal pressure hydrocephalus and multiple sclerosis, and pathologies associated with altered cerebral autoregulation such as traumatic brain injury.
	Our study ended in September, 2019. In the astronaut population, we demonstrated important, clinically relevant findings, which were previously unsuspected, concerning the brain health of long-duration astronauts and the process of intracranial adaption to spaceflight. Our work has resulted in significant progress towards closing gaps related to understanding the risks of CNS associated performance decrements and Spaceflight Associated Neuro-ocular Syndrome (SANS). We first reported: (1.) the development of ventricular enlargement correlated with mission duration in ISS astronauts (Roberts et al., N Engl J Med. 2017), (2.) microgravity-induced shifts in global brain structure (Roberts et al., N Engl J Med. 2017), and (3.) the correlation between brain structural changes and post-flight performance on standard cognitive and motor tests (Roberts et al., AJNR 2019). We also first reported that astronauts who developed SANS demonstrated minimal change in ventricular volumes while astronauts without SANS were found to have increased ventricular volumes following spaceflight on the International Space Station (ISS) (Roberts et al., AJNR, 2019). Based on these and our prior results, we hypothesis that the development of SANS may be related to human brain elasticity. We believe it is important to explore this association further. We have also correlated our quantitative results with changes in motor performance (Dr. Bloomberg's functional task testing data) and cognitive testing (WinSCAT) (Roberts et al., AJNR, 2019). We found that long-duration spaceflight resulted in significant crowding of brain parenchyma at the vertex. We identified specific locations in the brain which undergo structural change associated with changes in performance. Pre- to post-flight structural change of the left caudate correlated significantly with poor postural control and right primary motor area/mid cingulate correlated significantly with a complex motor task completion times. Change in volume of three white matter regions involving the visual pathways significantly correlated with altered reaction times on a cognitive performance task (bilateral optic radiations, splenium of the corpus callosum). Our results provide important insight into the adaptation of the human central nervous system to spaceflight and have important implications for planning human missions to the Moon and Mars. We have shown that prolonged exposure to

Task Progress:	microgravity is associated with brain structure, cognitive, and motor performance changes. Our results suggest that changes in ventricular volume and local brain structural changes have behavioral consequences in astronauts. Our findings support the need for advanced neuroimaging protocols and long-term follow-up imaging of the astronaut population (Roberts and Petersen, JAMA Neurology, 2019). We found that long-duration spaceflight was associated with the development of mastoid air cell effusions which was previously unsuspected. However, surprisingly no significant change in paranasal sinus opacification was noted (Inglesby et al., JAMA Otolaryngology, in press). This result has implications for engineers designing pressurized habitats and EVA (extravehicular activity) protocols that protect against barotrauma and for medications that might be considered for an exploration class mission. Most importantly, we have raised awareness concerning the intracranial findings associated with long-duration spaceflight. Addressing these changes will ultimately improve the health and safety of long-duration astronauts both on-board the International Space Station and those astronauts who will participate in future exploration class missions to the Moon and Mars. We are completing our final data analysis and preparing manuscripts under a No-Cost extension. We are continuing our collaboration with another investigator (Seidler) in the jointly sponsored NASA-German space agency bedrest study performed at the Envihab facility in Cologne, Germany. We have analyzed data concerning the physiological changes of the brain following long-term head down tilt bedrest in a CO2 environment as an analog for spaceflight. We are currently preparing the manuscript describing our data. Our results were quite surprising. We had hypothesized that head down tilt bedrest in combination with elevated CO2 would result in an increase in cerebral blood flow. Instead we found the opposite – there was a global decrease in cerebral blood flow compared with baseline. However, in subjects who developed SANS, gray matter perfusion was higher at day 29 of bedrest than in subjects who did not. We are now further collaborating with Dr. Seidler to determine the significance of this result. Our project team participated in the annual NASA Human Research Program Investigators Workshops held in Galveston (January, 2020). As an outreach effort, Dr. Roberts participated in the South Carolina Space Grant Consortium Summer Research Program, by mentoring four summer students (1 undergraduate, 1 medical student, 2 high school students) who assisted with analysis of data for this study. [Ed. note 2/26/2020: see Cumulative Bibliography for text references citations]
Bibliography Type:	Description: (Last Updated: 01/15/2026)
Articles in Peer-reviewed Journals	Inglesby DC, Antonucci MU, Spampinato MV, Collins HR, Meyer TA, Schlosser RJ, Shimada K, Roberts DR. "Spaceflight-associated changes in the opacification of the paranasal sinuses and mastoid air cells in astronauts." JAMA Otolaryngol Head Neck Surg. 2020 Jun 1;146(6):571–7. Published online Mar 26, 2020. https://doi.org/10.1001/jamaoto.2020.0228 ; PMID: 32215610; PMCID: PMC7099529 , Jun-2020
Articles in Peer-reviewed Journals	Roberts DR, Asemani D, Nietert PJ, Inglesby DC, Eckert MA, Bloomberg JJ, George MS, Brown TR. "Prolonged microgravity affects human brain structure and function." AJNR Am J Neuroradiol. 2019 Nov;40(11):1878-85. Epub 2019 Oct 17. https://doi.org/10.3174/ajnr.A6249 ; PubMed PMID: 31624117; PubMed Central PMCID: PMC6975111 , Nov-2019
Significant Media Coverage	Roberts D. "2019 TedX White Point Gardens Women, featured speaker, 'Gravity is Good for the Brain.' " TedX talk, December 7, 2019. https://www.ted.com/talks/donna_roberts_gravity_is_good_for_the_brain , Dec-2019
Significant Media Coverage	Talan J. "Prolonged Space Flight Affects Human Brain Structure and Function. Dr. Roberts is interviewed about her research." Neurology Today, December 5, 2019. https://journals.lww.com/neurotodayonline/Fulltext/2019/12050/Prolonged_Space_Flight_Affects_Human_Brain.2.aspx , Dec-2019
Significant Media Coverage	Mathewson S. "Long Space Missions Can Change Astronaut Brain Structure and Function. PI Roberts interviewed about her research." Space.com site, November 15, 2019. https://www.space.com/spaceflight-changes-astronaut-brain-structure-function.html , Nov-2019
Significant Media Coverage	O'Connor M. "Brain MRI scans may help NASA plan safer missions to space. Dr. Roberts is interviewed discussing her research." HealthImaging News, October 29, 2019. https://www.healthimaging.com/topics/advanced-visualization/brain-mri-help-nasa-plan-safer-missions-space , Oct-2019